Summable fourth-norm tails for the complete singlet bilocal

Common-disorder finite-time four-point functions in complex DSSYK

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7 November 2026 Local research draft

Abstract

We close the finite-time fourth-norm tail left open by the preceding bilocal reconstruction. Restricted Gaussian covariances and an ancilla contraction bound control pure marked words at every degree. An exact sequence of completely positive charge-transfer maps then controls the actual neutral covariance residual. Clifford Riesz inequalities, charge finite differences and an inverse Jacobi expansion give the corresponding bound for ordinary charged powers. The resulting tails are uniform at finite size. Together with the previously established same-disorder strong replacements and return estimate, they construct the full finite-time bilocal four-point function by convergent finite pairing sums. The result concerns the specified microscopic model and tracial state; the state/polarization and bulk-generator dictionary are the next physical questions.

1 Objects and result

Use standard CAR, $\{c_i, c_j^\dagger\} = \delta_{ij}$, and

$$H_N = \sqrt{w_N} \sum_{|I|=|J|=p} J_{IJ} C_I^\dagger C_J, \qquad w_N = \frac{4^p J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2. \tag{1}$$

The Gaussian matrix J is Hermitian, with $\mathbb{E} J_{IJ} J_{KL} = \delta_{IL}\delta_{JK}$; I, J may overlap. Every sector uses this same J. Set $\tau_N = \mathbb{E}_J 2^{-N} \operatorname{Tr}$ and $\|X\|_r = (\tau_N |X|^r)^{1/r}$. The state is $\beta = \mu = 0$. Take even N, $p \to \infty$, $\lambda_N = p^2/N \to \lambda > 0$ in a positive compact set, fixed J_s, $\hbar = 1$, and string time. All norm orders and real time windows below are fixed.

For neutral operators define $D = Q - \sum_i c_i^\dagger(\cdot) c_i$ and the half-Clifford length $\mathcal{L}\Gamma_A = |A|\Gamma_A/2$. $\mathcal{N}$ and $\mathcal{W}_j$ instead denote coupling Gaussian number and chaos projection. Write $Z_j = \mathcal{W}_j(H^j)$ and $C = J_s^2/\lambda$. The actual bilocal is the previously defined $\mathcal{B}_N(t) = (1 - \mathcal{P}_H) m e^{-iH_m t/2} F_m(t) e^{-iH_m t/2}$, where $F_m(t) = m^{-1} \sum_i c_i^\dagger e^{itH_{m-1}} c_i$; inactive sectors have zero dynamic field. Pinching $\mathcal{P}_H$ retains whole degenerate energy blocks.

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Theorem 1 (Finite-time four-point construction)**.** *The charged and neutral series in the two-component memory reduction have uniformly summable tails in every fixed finite S_r norm, $r \geq 2$. In particular their degree-n building blocks admit bounds of the form*

$$\|X_{n,N}\|_r \leq K_r(1+n)^{b_r} L_r^n \sqrt{n!}, \tag{2}$$

with constants independent of N, n. Consequently the joint ordered four-point functions of $\mathcal{B}_N(t)$ and its adjoint, with arbitrary fixed center-time conjugations, converge locally uniformly in all these real times. Their limit is the convergent pairing construction in Section 10.

The complete compressed-bath memory is connected by an S_r strong approximation; no separate claim about its own Taylor tail is needed. This is a model theorem assembled from the new estimates below and the explicit preceding strong interfaces [3, 6, 7]. It does not assert a uniform growing-time limit or interchange the double-scaling limit with $\lambda \downarrow 0$.

2 Restricted covariances without absolute CAR weights

Let $A_{IJ} = \sqrt{w_N} C_I^\dagger C_J$. On charge m,

$$\mathbb{E} H_m^2 = C_m I, \qquad C_m = w_N \binom{m}{p} \binom{N-m+p}{p} \leq C_*, \quad C_* := \frac{J_s^2}{\lambda_-} e^{2\lambda_+}. \tag{3}$$

Impose finitely many site requirements $T_I \subset I, T_J \subset J$. On occupation O, $|O| = m$, the diagonal of the column covariance vanishes unless $T_J \subset O$. Otherwise, with $K = T_J \cup (T_I \cap O)$, it equals

$$w_N \binom{m-|K|}{p-|K|} \binom{N-m+p-|T_I|}{p-|T_I|}. \tag{4}$$

Invalid binomial coefficients are zero. Thus, on an interior charge interval, with $t = |T_I| + |T_J|$ fixed, the row and column covariances are bounded by

$$C_m \frac{(p)_{|T_J|}}{(m)_{|T_J|}} \frac{(p)_{|T_I|}}{(N-m+p)_{|T_I|}} \leq C_* (K_t p/N)^t. \tag{5}$$

The row version swaps the two label roles.

For a fixed Gaussian pairing, scan the matrix word from right to left. At the first endpoint of an index use the column map $V\psi = \sum_\alpha A_\alpha \psi \otimes e_\alpha$; at its second endpoint use the row map. Their norms are the square roots of the corresponding covariance norms. Open indices occupy distinct ancillas; crossings only permute these ancillas unitarily. Fixed CAR sandwiches are contractions. The complete pairing contraction is therefore bounded by the product of endpoint norms. Requirements at both endpoints are first combined into the same allowed index family. This argument retains the CAR signs until the matrix contraction is bounded. Taking absolute local weights first would lose a factor exponential in the square of the degree.

3 Pure marks, repeated deficits and Gaussian projection

The exact body product rule is

$$D(XY) = (DX)Y + X(DY) - \sum_i [c_i^\dagger, X][Y, c_i]. \tag{6}$$

Since $DA_{IJ} = pA_{IJ}$, a deficit $(np - D)$ on a product of n slots selects two slots and one site. The two commutators require $i \in J_a$ and $i \in I_b$. In a pure Wick block those slots cannot pair with each other, so these are two independent set labels. The single-slot part of $p - \mathcal{L}$ requires $i \in I \cap J$ and has the same property. For s distinct sites,

$$N^s (p/N)^{2s} = \lambda_N^s. \tag{7}$$

Combining this count with (5), a word with L slots and k fixed pure marks satisfies

$$|\tau_N(\text{marked word})| \le K_k (1 + L)^{2k} C_*^{L/2} (L - 1)!!. \tag{8}$$

The outside-charge estimate uses at worst N^k and the binomial tail; its exponential decay absorbs this fixed power. For four copies of a one-mark degree-n field, $(4n - 1)!! \le 16^n (n!)^2$, giving (2).

Repeated deficits on one pure block also obey this bound. Extend the body operator to a homogeneous parity-a word by

$$D_g X = \tfrac{1}{2}\{Q, X\} - (-1)^a \sum_i c_i^\dagger X c_i. \tag{9}$$

A normal ordered word with $r_c + r_a$ letters has eigenvalue $(r_c + r_a)/2$. Each contraction removes two letters. Applying the next deficit either counts the already removed pairs or adds another tagged contraction. The multiplicity depends on the fixed number of marks, with no new p loss. Therefore any fixed sequence of k $D/\mathcal{L}$ deficits has bound $K_{k,r}(1 + n)^{2k} L_{k,r}^n \sqrt{n!}$.

There is also a dimension independent, same-norm Gaussian tool. For $\deg P \le n$, the contraction $\mathsf{O}_t P = \sum_{j \le n} t^j \mathcal{W}_j P$, $-1 \le t \le 1$, has a Chebyshev expansion whose coefficients have norm at most $2\|P\|_r$. Leading coefficients and absolute coefficient sums give

$$\|\mathcal{W}_n P\|_r \le 2^n \|P\|_r, \tag{10}$$

$$\left\| \sum_{j \le n} \theta_j \mathcal{W}_j P \right\|_r \le K (1 + \sqrt{2})^n \max_j |\theta_j| \, \|P\|_r. \tag{11}$$

Linear Gaussian matrix moments [2] yield $\|H_m^n\|_r \le K_r^n \sqrt{n!}$; hence the same envelope holds for Z_n. No matrix-valued S_2 to S_4 hypercontractivity is assumed.

4 The covariance is an exact completely positive path

On $\mathcal{H}_k = \wedge^k \mathbb{C}^N$ define

$$T_k X = \frac{1}{k} \sum_i c_i^\dagger X c_i, \qquad S_k X = \frac{1}{N-k} \sum_i c_i X c_i^\dagger. \tag{12}$$

T_k maps charge $k-1$ to k and S_k maps $k+1$ to k. Both are unital and preserve their respective normalized sector traces. Thus they contract every Schatten norm, also after disorder averaging. The deterministic covariance map has the exact factorization

$$\Phi_m(X) = w_N \sum_{I,J} C_I^\dagger C_J X C_J^\dagger C_I, \qquad \frac{\Phi_m}{C_m} = T_m \cdots T_{m-p+1} S_{m-p} \cdots S_{m-1}. \tag{13}$$

Expanding the Kraus products proves this identity: each unordered p-set has $p!$ orders on each side, the reordering signs cancel, and the path denominators give $(m)_p (N-m+p)_p = (p!)^2 \binom{m}{p} \binom{N-m+p}{p}$.

Put $\bar{D} = 2\mathcal{L} - D$. For any neutral full-Fock section X,

$$T_k X_{k-1} - (1 - b/k) X_k = ((b - D)X)_k / k, \tag{14}$$

$$S_k X_{k+1} - (1 - b/(N-k)) X_k = ((b - \bar{D})X)_k / (N-k). \tag{15}$$

Set

$$\theta_{v,m} = \prod_{a=0}^{p-1} (1 - vp/(m-a))_+ (1 - vp/(N-m+p-a))_+, \tag{16}$$

$$A_{v,m} = p(\Phi_m - C_m \theta_{v,m}) Z_{v,m}. \tag{17}$$

Inactive sectors are defined separately by zero covariance. In $N/3 \le m \le 2N/3$, the path stays in a wider interior interval. If $vp \le N/8$, telescope (14) over $2p$ steps. Every source has coefficient at most Kp/N after multiplying by pC_m; their sum costs $p^2/N = \lambda_N$. The pure marked bound therefore proves (2) for A_v. If $vp > N/8$, then $p < 8\lambda_N v$, and the coarse CP estimate $\|A_v\|_r \le 2pC_* \|Z_v\|_r$ suffices. Outside the central interval the same estimate is absorbed by the binomial tail.

This supplies a bound for the actual covariance residual, before taking any degree or size limit. A coefficientwise occupation Markov estimate alone would not control its summed Gaussian Schatten norm; the full CP factorization is the needed bridge.

5 The actual neutral time series

Retain finite-sector parameters in the Jacobi recurrence:

$$a_{j,m} = \sum_{v<j} \theta_{v,m}, \qquad Q_{j+1}(x) = xQ_j(x) - C_m a_{j,m} Q_{j-1}(x), \qquad \eta_j = p[Z_j - Q_j(H)]. \tag{18}$$

The exact Gaussian creation/annihilation identity from the previous neutral stage [4] is

$$pr_j = \sum_{v+w=j-1} A_v \diamond Z_w, \qquad \eta_{j+1} = H\eta_j - C_m a_j \eta_{j-1} - pr_j. \tag{19}$$

Since A_v is pure coupling degree v,

$$A_v \diamond Z_w = \mathcal{W}_{v+w}(A_v H^w). \tag{20}$$

Equation (10) and fixed $2r$ Hölder give an exponential times $\sqrt{(v+w)!}$ envelope. Unroll (19) before taking norms. Its terms are bounded Jacobi path coefficients times $H^a \mathcal{W}_{v+w}(A_v H^w)$. Only a fixed $4r$ input is needed, independent of the degree. The path count is exponential and $\sqrt{a!v!w!} \le \sqrt{j!}$, giving (2) for η_j. Outer charges can instead use $p(Z_j - Q_j(H))$ and the binomial tail.

The exact finite-parameter expansion $e^{isH_m} = \sum_j c_{j,m}(s) Q_{j,m}(H_m)$ obeys $|c_{j,m}(s)| \le K_S L_S^j / j!$. Subtracting a function of H_m and using exact pinching gives

$$p(1-\mathcal{P}_H)\Phi_m(e^{isH_m}) = (1-\mathcal{P}_H)\sum_{j\ge 0} c_{j,m}(s) T_{j,m}, \tag{21}$$

$$T_{j,m} = A_{j,m} + (C_m \theta_{j,m} - \Phi_m)\eta_{j,m}. \tag{22}$$

The normalized covariance is a contraction, so T_j satisfies (2). With $\alpha_m = 1 - p/m$ and $\epsilon_m = 1 - \alpha_m^2$, the actual neutral memory is

$$B_{m,N}(s) = \frac{m\epsilon_m}{p}(1-\mathcal{P}_H) e^{-iH_m s/2} \sum_j c_{j,m}(\alpha_m s) T_{j,m} e^{-iH_m s/2}. \tag{23}$$

Here $0 \le m\epsilon_m/p \le 2$ on active sectors, and the phases are unitary. Thus the actual remainder, uniformly at finite N, is bounded by

$$K_{r,S} \sum_{j>K} (j+1)^{b_r} L_{r,S}^j / \sqrt{j!} \longrightarrow 0. \tag{24}$$

The previously constructed local-size/ordered-Wick proxy has the same bound by Section 3. The old compact-time L^2 strong bridge therefore upgrades by interpolation to every fixed S_r.

6 Differentiating the CP path without a size loss

Use distinct notation for unnormalized maps $K_-X = \sum c_i^\dagger X c_i$, $K_+X = \sum c_i X c_i^\dagger$, and normalized maps $t = Q^{-1}K_-$, $s = (N-Q)^{-1}K_+$. Direct CAR give

$$[D,t] = t - t^2, \qquad [\bar{D},s] = s - s^2, \tag{25}$$

$$[D,s] = -s + \frac{Q}{N-Q}ts - \frac{2Q-N}{N-Q}I, \tag{26}$$

$$[\bar{D},t] = -t + \frac{N-Q}{Q}st + \frac{2Q-N}{Q}I. \tag{27}$$

For a scalar charge function, retain the finite differences:

$$[D, f(Q)] = [f(Q) - f(Q-1)]K_-, \tag{28}$$

$$[\bar{D}, f(Q)] = [f(Q) - f(Q+1)]K_+. \tag{29}$$

Commuting D/p or $\mathcal{L}/p$ through at most $2p$ normalized steps costs only a bounded factor on the interior interval. Apply this to the telescope for A_v, not to its coarse difference. Each term is $pC(Q)f_h(Q)P_hE_h$, with a pure deficit E_h, $|f_h| \le K/N$, and $|\Delta f_h| \le K(1+v\lambda_N)/N^2$. Repeated pure marks control derivatives of E_h; (28) controls all scalar coefficients. For $vp > N/8$, use $p \le 8\lambda_+ v$ and $N \le 64\lambda_+ v^2$ instead. Outer charges are controlled before applying any full-trace theorem. The result is

$$\|A_v\|_r + \|DA_v/p\|_r + \|\mathcal{L}A_v/p\|_r \le K_r(v+1)^{b_r}L_r^v\sqrt{v!}. \tag{30}$$

The Clifford Riesz theorem of Ben Efraim–Lust-Piquard, Theorem 6.8(1),(3), and its semigroup interpolation Lemma 5.6 [1], supply the product bridge. In the present convention $N_{\text{Clifford}} = 2\mathcal{L}$. For either physical product-rule cross term $\Gamma_\sigma(X,Y)$, $\sigma = D, \mathcal{L}$,

$$\|\Gamma_\sigma(X,Y)/p\|_r \le K_r\big[\|X\|_{2r}\|\mathcal{L}X/p\|_{2r}\|Y\|_{2r}\|\mathcal{L}Y/p\|_{2r}\big]^{1/2}. \tag{31}$$

These are normalized annealed *full-Fock* norms. The theorem is not applied within a fixed charge sector. Its row and column square functions must both be kept for non-Hermitian inputs.

7 Ordinary derivatives and the Jacobi error

A useful independent CP estimate avoids auxiliary dimension factors. If Φ is completely positive, with trace adjoint $\Phi^\dagger$, then scalar Hölder on the spectral decompositions of positive X, Y gives, for conjugate u, v,

$$\operatorname{tr} Y\Phi(X) \le [\operatorname{tr} X^u \Phi^\dagger(I)]^{1/u} [\operatorname{tr} Y^v \Phi(I)]^{1/v}. \tag{32}$$

Disorder can be included in that same Hölder inequality. Duality with $u = 2r$ then gives

$$\|\Phi(X)\|_r \le \|X\|_{4r} \|\Phi^\dagger(I)\|_2^{1/(2r)} \|\Phi(I)\|_{2r-1}^{1-1/(2r)}. \tag{33}$$

For arbitrary X an absolute factor at most four suffices. For the force map

$$\Gamma_H(X) = \sum_i [c_i^\dagger, H] X [H, c_i], \tag{34}$$

the Clifford square-function bound controls both Grams by $O_r(p)$. Consequently $\|\Gamma_H(X)\|_r \le K_r p \|X\|_{4r}$. The exact ordered identity

$$DH^w = pwH^w - \sum_{a+b+c=w-2} H^a \Gamma_H(H^b) H^c \tag{35}$$

and the analogous $\mathcal{L}$ formula imply

$$\|H^w\|_r + \|DH^w/p\|_r + \|\mathcal{L}H^w/p\|_r \le K_r (w+1)^{b_r} L_r^w \sqrt{w!}. \tag{36}$$

Alternatively, the tagged proof gives this independently: ordinary internal pairs may leave only one independent set requirement per site, but each normalized deficit supplies $1/p$. For $s \le k$ distinct sites in k marks, $p^{-k} N^s (p/N)^s = p^{s-k} \le 1$.

Now unroll (19). Its terms have the form

$$b(Q) H^a \mathcal{W}_{v+w}(A_v H^w). \tag{37}$$

Use (31) once inside the projection and once outside it; physical derivatives commute with Gaussian projection. The needed norm order is a fixed multiple of r, not one that grows at each recursive step. Coefficient derivatives use (28) separately. The central Jacobi path weights and their normalized differences grow at most exponentially times a degree polynomial. The outer estimate follows directly from $p(Z_j - Q_j(H))$ and its derivatives. It follows that

$$\|\eta_j\|_r + \|D\eta_j/p\|_r + \|\mathcal{L}\eta_j/p\|_r \le K_r (j+1)^{b_r} L_r^j \sqrt{j!}. \tag{38}$$

8 The ordinary charged tail: keep the charge differences

The inverse Jacobi expansion is exact:

$$H^n = \sum_{j=0}^{n} h_{nj}(Q)[Z_j - \eta_j/p]. \tag{39}$$

Its coefficients are weighted up/down paths. On an interior charge interval the recurrence weights $\beta_j = C_m \sum_{v<j} \theta_v$ are bounded. The sharper difference estimate needed here follows from

$$\frac{C_m}{C_{m-1}} = \frac{m}{m-p}\frac{N-m+1}{N-m+p+1}, \qquad N|\Delta C_m| \le K\lambda_N(1+|X_m|), \quad X_m = \frac{m-N/2}{p}. \tag{40}$$

Each positive-part factor in θ_v is Lipschitz, including across its cutoff. All central denominators are at least cN, so

$$N|\Delta\theta_{v,m}| \le K\lambda_N v \quad \text{for every } v. \tag{41}$$

Taking one difference along each path, rather than differentiating every coefficient simultaneously, gives

$$\sum_j |h_{nj,m}| \le K^n, \qquad \sum_j N|\Delta h_{nj,m}| \le K(n+1)^3 L^n(1+|X_m|). \tag{42}$$

Binomial full-Fock charge has bounded fixed moments of X_m.

Define the actual ordinary charged coefficient

$$G_{n,N} = [p\mathcal{N} - D - \lambda_N \mathcal{N}(\mathcal{N}-1)]H^n. \tag{43}$$

With $\Delta h_m = h_m - h_{m-1}$, its non-counterterm part is

$$\begin{aligned}(p\mathcal{N} - D)H^n = \sum_j \big[&h_{nj}(pj - D)Z_j - (\Delta h_{nj})K_- Z_j - h_{nj}\mathcal{N}\eta_j \\ &+ h_{nj}D\eta_j/p + (\Delta h_{nj})K_-\eta_j/p\big].\end{aligned} \tag{44}$$

This retains all internally contracted Gaussian terms. The first term is pure marked; the second uses (42); the third uses (11); the last two use (38) and coefficient differences. On outer charges use the original G_n and the coarse $D = O(N)$ bound with exponential binomial tails. Finally $\lambda_N\mathcal{N}(\mathcal{N}-1)$ is controlled by (11). Thus (2) holds for G_n and for

$$K_n = G_{n+2} - \{H, G_{n+1}\} + HG_nH. \tag{45}$$

The actual series $\sum_n (is)^n G_n/n!$ and its fixed derivatives and energy insertions therefore have the finite-N tail (24). This supplies the missing ordinary, rather than only pure-Wick, bound.

9 Returning to the actual fields

The preceding memory reduction [5, 6] is also valid in every fixed S_r norm. Here is the required upgrade, specifying which bounds are used.

First, the charged OU remainder is exactly

$$E_m - G_N = R_m(\mathcal{N})e^{isH}, \qquad E_m = [m(1-\alpha_m^{\mathcal{N}}) - D]e^{isH}|_m. \tag{46}$$

The physical derivative cancels. The central bound $|R_m(k)| \le a_m(k+1)^3$, with $\|a_m\|_{L^r(\text{binomial})} = O_r(p^{-1})$, combines with (11) and the ordinary moment envelope to give $O(p^{-1})$ for fixed derivatives and energy insertions. The other $\alpha_m - 1$ terms vanish by the new charged bounds (or by the earlier coarse high-moment interpolation).

Second, the old positive-force estimates bound $p\Lambda(s)$ in every fixed norm. Its L^2 convergence to the energy regression therefore upgrades by interpolation. Fixed-order Hölder transfers this to the first return; the next return is $O(p^{-1})$. The four-auxiliary-Gaussian energy regression and the next OU remainder are treated at fixed higher norm orders in the same way. This gives the actual memory to charged-plus-neutral strong bridge.

Third, for each fixed n, the joint-regression result uses the same original disorder:

$$R_{n,N} := G_{n,N} - g_n(X_N, H_N) \longrightarrow \sum_d a_{nd} J_{d,N} \quad \text{strongly in } L^2. \tag{47}$$

The new G_n bounds, the fixed-degree charge/energy polynomial bounds, and pure marked bounds upgrade this to S_r. Ordinary internal pairs are removed through (47); they are not silently treated as forbidden pure-Wick pairs.

The neutral local proxy is a finite sum of ordered pure marked blocks and ordinary outside powers at each fixed degree. Its previous strong bridge and the new two-sided high bounds give the same upgrade. Fixed residuals have no diagonal spectral mass by the preceding two-point argument. Exact pinching first vanishes in L^2, then in S_r by interpolation and contraction. Only then are the finite-degree common-source moments used.

Here $\mathcal{F}_N(s)$ is the actual centered memory field: $m(1-\mathcal{P}_H)e^{-iH_m s/2}M_m(s)e^{-iH_m s/2}$, with the compressed-bath memory M_m of the preceding reconstruction. The established bilocal return estimate [3] is

$$\sup_{|t|\le S} \|\mathcal{B}_N(t) - \widetilde{\mathcal{B}}_N(t)\|_r \le K_{r,S}p^{-1/(2r-1)} + K_{r,S}pe^{-N/(8r)}, \tag{48}$$

$$\widetilde{\mathcal{B}}_N(t) = -\int_0^t ds \int_{-(t-s)/2}^{(t-s)/2} du\, e^{iH_N u}\mathcal{F}_N(s)e^{-iH_N u}, \quad t \ge 0. \tag{49}$$

At $r = 4$ the return error is $O(p^{-1/7})$.

10 A constructive complete four-point function

Let τ_∞ be the limiting common-source tracial functional. At fixed degree it is explicitly evaluated as follows. Expand all fields and counterterms into ordinary outside powers and pure marked blocks; keep the whole-block Wick restrictions and the true matrix order. For every allowed pairing, form the local two-label CAR polynomial $T_{uv}(z)$ and divide by its source-dependent singleton factors, obtaining $R_{uv}(z)$. Extract the required nilpotent source coefficient from

$$(Ce^{-\lambda})^\ell \exp\left\{\lambda \sum_{u<v} [R_{uv}(z) - 1]\right\}, \tag{50}$$

then sum the allowed pairings. Counterterms and chaos projectors are expanded with their actual coefficients. The prefactor follows from $w_N \binom{N}{p}^2 2^{-2p} \to Ce^{-\lambda}$ per chord. Finite-size overlap exponential moments justify this fixed-word Poisson limit [3]; Section 3, separately, gives summable degree bounds. General descendants are not replaced by the three-species low-field mixed-q law.

Write $R_n^\infty = \sum_d a_{nd} J_d$ and $L_n = R_{n+2}^\infty - \{H, R_{n+1}^\infty\} + H R_n^\infty H$. Let N_j be the centered neutral ordered descendant. For $q = e^{-4\lambda}$, let H_j be the monic q-Hermite polynomial of squared norm $C^j[j]_q!$ in ρ_q, and put $c_j(s) = \int e^{isx} H_j(x) d\rho_q(x)/(C^j[j]_q!)$. Define

$$\mathcal{F}^{[K]}(s) = e^{-iHs/2} \left[\sum_{n=0}^{K} \frac{(is)^n}{n!} L_n + 2C \sum_{j=1}^{K} c_j(s) N_j\right] e^{-iHs/2}. \tag{51}$$

The neutral sign is positive. Apply the filter in (49) to define $\mathcal{B}^{[K]}(t)$, and extend to negative times by adjoints. Fixed-degree strong replacement and $\|1 - \mathcal{P}_H\|_r \le 2$ make the limiting centered blocks inherit the degree envelope; this does not rely on L^2 regression being an S_r contraction. For $\varepsilon_i \in \{1, *\}$, center times T_i, and relative times t_i, the answer is

$$\boxed{\lim_{K\to\infty} \tau_\infty \prod_{i=1}^{4} \left(e^{iHT_i} \mathcal{B}^{[K]}(t_i) e^{-iHT_i}\right)^{\varepsilon_i}.} \tag{52}$$

At fixed K, expand the energy exponentials into polynomials and evaluate them by (50); Gaussian energy moments control this auxiliary truncation. Thus (52) is a convergent algorithm, without requiring a closed four-field spectral kernel.

For completeness, the filter norm is at most $S^2/2$. The new S_4 tails make its truncated fields Cauchy, and all actual fields are bounded in S_4. Telescoping a four-factor product and using Hölder bounds its error by one S_4 difference times three S_4 bounds. First pass $N \to \infty$ at fixed truncation using the strong interfaces; then remove truncation using (24). The return contribution is controlled by (48). All these steps are locally uniform in the fixed time windows. Unitary insertions preserve strong-error bounds for arbitrary center times, but spectral convergence itself is only asserted on compact windows.

11 Checks, provenance and the next physical step

Three independent finite algebra checks accompany the proof. They test exact identities and nontrivial CAR signs; they do not substitute for the all-degree estimates.

- Restricted covariance and ancilla contraction: 42,336 covariance comparisons, 564 networks (452 crossed), 94,743 label assignments, and 1,776 exact rational sector bounds. There are 113 nonzero crossed CAR-sandwich cases.
- Coherence/CP channel: 340 coherence identities, 265 normalized path identities, 75 inactive cases, 2,609 kernel entries and 1,305 tagged-site bounds, including 132 nonconstant inputs and 14 nontrivial CAR gauges.
- Charged reassembly: 3,312 matrix identities, 9,949 entry comparisons and 84 polynomial identities, including 16 nonzero charge-difference fixtures. The polynomial fixtures use one Gaussian coordinate to check algebra, not the full ensemble's moments.

Drivers and outputs are stored under `code/` and `results/susskind-bilocal-fourth-tail/`. The local evidence manifest binds the input/output hashes, parameters, dependency versions, parent manifests, build logs and final page renders. The two preceding checks were first successful runs; old scientific outputs and failed records are preserved. The new third check is also saved independently, rather than overwriting a predecessor.

The theoretical increment is precise: complete finite-relative-time four-point functions now follow from actual finite-size uniform tails, rather than from a formal limiting Taylor series. The next question is which state and polarization can turn this tracial collective algebra into a physical particle construction, and how the microscopic Hamiltonian supplies spatial/internal generators. The positive low-channel commutator remains a useful calibration. It is not by itself a finite-G gravitational scattering amplitude. No new issue reply, Git commit or push accompanies this local stage.

References

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[6] A. Day and Chengheng, *Joint charge/energy regression and the complete memory kernel*, local draft and proof notes (7 Nov. 2026).

[7] A. Day and Chengheng, *Strong reconstruction of the complete bilocal fluctuation*, local draft (7 Nov. 2026).